Two-Stream Spatial–Temporal Fusion for River Water Quality Prediction with Anomaly Screening and Source Shortlisting

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***Abstract*—A purely autoregressive water quality model can learn when parameters move but not how far, because amplitude is set by catchment conditions absent from a station's own history. This paper presents a two-stream architecture that supplies the missing information. A temporal expert applies CEEMDAN decomposition followed by a CNN–BiLSTM with self-attention; a spatial expert derives 200 features from Sentinel-2 and Landsat-8 imagery over each station's catchment, selects a compact subset by evolutionary search, and regresses them with a random forest. An attention gate learns per-sample weights over the two experts. Evaluated on 4,048 daily records from seven Danube stations (2012–2023), fusion raises R^2 to 0.612 for nitrate-N, 0.459 for dissolved oxygen and 0.410 for electrical conductance, against 0.517, 0.322 and 0.312 for the temporal stream alone. We further compare four hyperparameter and feature selection strategies, finding CMA-ES feature selection improves mean fused R^2 by 21% over a genetic algorithm while reducing tuning time by 15%. A downstream layer screens anomalies by dual rolling-median and seasonal tests, bounds the search region to a river stretch by cross-station comparison, and shortlists candidate industrial sites for investigation.**

***Index Terms*—Water quality prediction, sensor fusion, CEEMDAN, BiLSTM, random forest, CMA-ES, remote sensing, anomaly screening**

I. Introduction

Predicting river water quality is difficult because measurements are nonlinear, nonstationary, noisy and sparsely sampled. Phase 1 of this project addressed the temporal side of that problem with a CEEMDAN–CNN–LSTM model incorporating self-attention, and produced a diagnostic result that determined the direction of the present work. That model tracked seasonally driven parameters in correct phase but reproduced only 65–73% of their observed variance, and recovered no structure at all for parameters governed by episodic external forcing. In short, it learned *when* parameters move but not *how far*.

The interpretation is that the missing information is not temporal but spatial. Amplitude and event timing in a river are governed largely by catchment conditions – rainfall and runoff, land use, upstream discharge, sediment transport – none of which appear in a station's own measurement history. Remote sensing supplies exactly this class of observation [15]–[17], and is complementary to what a sequence model already provides.

Phase 2 therefore builds a *spatial expert* over remotely sensed catchment features and fuses it with the temporal expert under a learned gate, following the mixture-of-experts principle [8] in which a gating network arbitrates between specialists per input. The system was rebuilt on the Danube, whose record is denser than the sparse gauge data of Phase 1 and whose seven Serbian stations lie along a single channel, permitting spatial reasoning. Our contributions are: a two-stream fusion architecture with an interpretable attention gate; a controlled comparison of four hyperparameter and feature-selection strategies; and an anomaly screening layer that bounds events to a river stretch and shortlists candidate industrial sites.

II. Related Work

Zhu et al. [1] decompose raw series with CEEMDAN before an LSTM–CNN network with self-attention, reporting strong accuracy for dissolved oxygen and pH; this motivated the temporal stream. Yan and Wang [2] model Yellow River Basin stations as graph nodes, combining graph convolution with reinforcement learning to capture spatial dependency between sites, and demonstrate that inter-station structure carries predictive signal that single-site models discard.

For the spatial side, reviews of remote sensing for water quality retrieval [15], [16] establish that catchment-scale spectral information carries signal about nutrient and sediment loading, and Pahlevan et al. [17] demonstrate operational chlorophyll-a retrieval from Sentinel-2 and Sentinel-3. Shu et al. [3] combine super-resolved imagery with field measurements for groundwater mapping, weighting parameters by fuzzy AHP. These establish satellite imagery as a viable predictor source but typically use it in isolation rather than fused with a temporal model.

On optimisation, random search is a strong hyperparameter baseline [12]; Optuna's TPE [13] models the objective density and BOHB [14] adds successive halving for efficiency. For feature selection, genetic algorithms [9], [10] are conventional, while CMA-ES [11] adapts a full covariance model of the search distribution, suiting correlated landscapes – a plausible description of a spectral feature space with strongly intercorrelated bands and indices. We test that in Section V. Across this literature, results are often reported without baselines or per-component ablations, making it hard to establish where accuracy originates; we report per-stream metrics throughout.

III. System Architecture

The system comprises four layers: acquisition and preprocessing, the two-stream predictor, the fusion gate, and anomaly screening.

A. Temporal Expert Stream

Twelve signals per station – streamflow, precipitation, water temperature and nine spectral bands – are decomposed by CEEMDAN [4] into intrinsic mode functions, isolating oscillatory modes from high-frequency noise to low-frequency trend:

$$X(t) = \sum_{k=1}^{N} \mathrm{IMF}_k(t) + r_N(t) \tag{1}$$

The resulting 134-channel representation over a 30-day window passes through a 1-D CNN for local feature extraction, a bidirectional LSTM [5] for longer-range dependency, and a self-attention layer [6] that performs learned weighted pooling over the sequence, so each historical step's contribution is explicitly weighted.

B. Spatial Expert Stream

For each station and date, nine spectral bands (Blue through SWIR2, including three red-edge bands) and sixteen derived indices are computed. Beyond standard indices such as NDWI, NDVI and NDTI, several are domain-specific: a clay index and particle-size proxy marking phosphorus-adsorbing soil runoff, a dilution proxy relating conductance to discharge, and a Garcia–Gordon dissolved oxygen saturation term giving the physical upper bound at measured temperature. Each of the 25 base features is summarised over a 30-day window by seven statistics, giving 200 spatial features.

This dimensionality far exceeds the available measurements, so a compact subset is selected by evolutionary search – either a genetic algorithm [9] with L_0 penalty on the number of selected features, or CMA-ES [11] over a continuous relaxation – and regressed with a random forest [7], which is well suited to the resulting small-sample, high-dimensional regime.

C. Attention Fusion Gate

Rather than averaging the two experts, a gating network computes per-sample weights from the concatenated temporal hidden state $\mathbf{h}_t$, spatial feature vector $\mathbf{f}_s$ and an encoded hydro-meteorological context $\mathbf{c}$:

$$[w_t, w_s] = \mathrm{softmax}\big(\mathbf{W}_2\,\mathrm{ReLU}(\mathbf{W}_1[\mathbf{h}_t;\mathbf{f}_s;\mathbf{c}])\big) \tag{2}$$

$$\hat{y} = w_t\,\hat{y}_{\text{temporal}} + w_s\,\hat{y}_{\text{spatial}} \tag{3}$$

Because $w_t + w_s = 1$ and both are produced per sample, the gate is directly interpretable: it reports which expert the model relied on for each prediction.

D. Anomaly Screening and Source Shortlisting

This layer is decision support: it narrows where an investigator should look and does not determine causation, for reasons Section V-C sets out.

Anomalies are screened on measured values rather than predictions, using two tests that must both fire: a rolling Z-score against a 90-day median and MAD, and a seasonal test against the same-month mean across all years. Requiring both suppresses false positives from seasonal extremes, and events are classified *acute* or *trend* by whether the rolling mean is itself elevated. Because the seven stations lie along one channel at known river kilometres, the most upstream station showing an anomaly bounds the search region to the stretch above it – a geometric constraint, independent of any assumption about cause.

Industrial facilities within that stretch are retrieved from an OpenStreetMap database via the Overpass API and filtered by pollutant compatibility, giving a shortlist of sites whose effluent profile could in principle produce the observed signature. Events exceeding $Z \geq 3.5$ are routed to a large language model, which receives the measurement, baselines, multi-station context and shortlist, and returns ranked hypotheses with stated confidence and follow-up checks. Proximity and pollutant compatibility justify inclusion in a shortlist, not a finding of discharge; outputs are investigative leads requiring field verification.

IV. Implementation

A. Dataset

The Danube dataset covers seven Serbian monitoring stations from Bezdan (km 1425) to Tekija, spanning 1 December 2012 to 31 December 2023 – 4,048 daily records per station. Water quality targets are dissolved oxygen, total phosphorus, nitrate-N, electrical conductance and chlorophyll-a, drawn from GFQA records at 125–180 samples per station per parameter. Satellite observations come from Sentinel-2 and Landsat-8; discharge, precipitation and water temperature provide hydro-meteorological context.

Two preprocessing issues required correction. Google Earth Engine exported dates in day-of-year form (e.g. `2013-04-118`), which parses silently but wrongly unless the third field is detected as a DOY when it exceeds 31. Landsat-8 lacks red-edge bands and encodes them as −9999; these were converted to missing and interpolated from Sentinel-2 rather than treated as valid reflectance.

Critically, water quality targets are used only on actual measurement dates. Interpolating targets to a daily grid inflates apparent performance catastrophically, since the model then learns the interpolator rather than the process; input features may be interpolated, but labels may not.

B. Training and Optimisation

Models are implemented in PyTorch with Metal Performance Shaders acceleration. Training uses a masked MSE loss so that missing targets contribute no gradient, with early

TABLE I

PER-STREAM R^2, RANDOM SEARCH + CMA-ES (135 TEST SAMPLES)

Parameter	Temporal	Spatial	Fused
Dissolved oxygen	0.322	**0.629**	0.459
Nitrate-N	0.517	0.384	**0.612**
Electrical conductance	0.312	0.395	**0.410**
Chlorophyll-a	0.129	−0.255	**0.285**
Total phosphorus	−0.121	0.004	−0.052

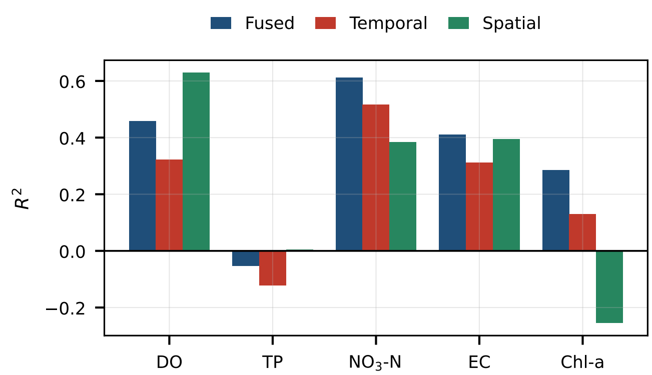

Fig. 1. Per-stream R^2 across the five targets. Fusion exceeds both individual experts for nitrate-N, conductance and chlorophyll-a, but not for dissolved oxygen, where the spatial stream alone is stronger.

stopping on validation loss. Splits are chronological, never random.

Four tuning strategies were compared under identical architecture and data, varying only how hyperparameters are chosen and how spatial features are selected: random search with GA selection (the baseline), Optuna TPE [13] with GA, BOHB [14] with GA, and random search with CMA-ES selection. Each used 3-fold cross-validation; results are reported on 135 held-out test samples.

V. RESULTS

A. Stream Contributions

Table I reports R^2 for each stream under the best configuration, and Fig. 1 shows the same comparison.

Fusion improves on the temporal stream for every parameter, confirming the Phase 1 hypothesis that spatial information supplies signal a sequence model cannot recover alone. Gains are substantial: nitrate-N rises from 0.517 to 0.612, conductance from 0.312 to 0.410, and chlorophyll-a from 0.129 to 0.285.

Two results deserve emphasis because they complicate the simple reading. For dissolved oxygen the spatial stream alone attains R^2 = 0.629, well above the fused 0.459 – the gate dilutes a strong expert with a weaker one. For chlorophyll-a the opposite holds: fusion (0.285) beats both temporal (0.129) and spatial (−0.255) streams, so the gate extracts value from an expert that is worse than useless on its own. Total phosphorus remains negative across all configurations, which is consistent with its measurement scale: values below 0.05 mg/L sit near the analytical detection limit, so a large fraction of the observed variance is measurement noise rather than signal.

The mean attention weights are 0.817 temporal and 0.184 spatial. The gate therefore leans heavily on the temporal expert overall, despite the spatial expert being the stronger predictor for dissolved oxygen and conductance. This imbalance, combined with the dissolved oxygen result, indicates the gate is under-trained relative to the streams it arbitrates and is the most promising target for further work.

TABLE II

TUNING STRATEGY COMPARISON (3-FOLD CV, 135 TEST SAMPLES)

HPO	Feat. sel.	Trials	Time (min)	Mean R^2
Random	GA	12	214	0.326
TPE	GA	20	318	0.365
BOHB	GA	22	197	0.360
Random	CMA-ES	12	**183**	**0.442**

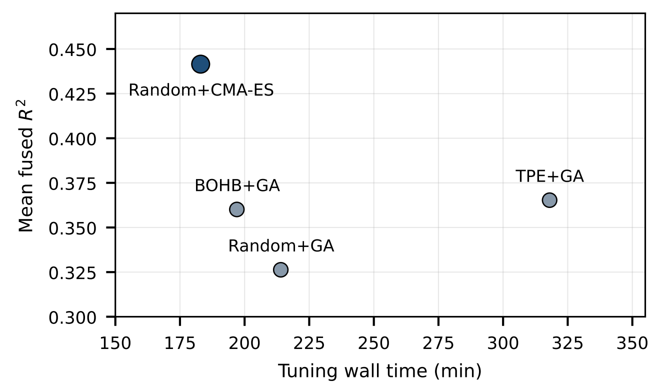

Fig. 2. Mean fused R^2 against tuning wall time. CMA-ES feature selection dominates: highest accuracy at the lowest cost.

B. Optimisation Strategy Comparison

Table II and Fig. 2 compare the four strategies.

The comparison isolates a clear effect. Changing the *hyperparameter* search from random to TPE or BOHB yields modest gains (0.326 to 0.365 and 0.360), at 49% more wall time for TPE. Changing the *feature selector* from GA to CMA-ES, while holding random search fixed, lifts mean fused R^2 to 0.442 – a 21% improvement over the best GA variant – while reducing tuning time to 183 minutes, the fastest of the four. Per-parameter gains over the GA baseline are +7.1% for dissolved oxygen, +10.3% for nitrate-N, +11.2% for conductance and +17.6% for chlorophyll-a. CMA-ES converged in 12.4 generations on average against 20 for the GA.

The magnitude of this result relative to the hyperparameter effect suggests that on this problem, *which spectral features enter the spatial expert* matters considerably more than how the network is configured. This is consistent with the intercorrelated structure of a spectral feature space, in which

CMA-ES's covariance model is better matched to the search landscape than the GA's independent bit-flips. We note the comparison rests on a single seed per configuration, so the ranking of TPE against BOHB is within plausible run-to-run variance; the CMA-ES margin is large enough to be robust to it.

C. Anomaly Screening: Scope and Evaluation Limits

On the historical record the screen flags events passing both the rolling and seasonal tests. A representative case is an electrical conductance reading of 387 μS/cm at Bogojevo on 10 October 2017 against a rolling median of 350 μS/cm, giving a rolling Z of 16.6 and classification as acute. With only that station affected, the search region is bounded to the 58 km stretch between Bezdan (km 1425) and Bogojevo (km 1367). No facility in the industrial database matched this stretch and pollutant, which the report records explicitly rather than returning a source.

We report no precision, recall or attribution accuracy here, because the required ground truth does not exist in our evaluation set. Validation would need dated incident logs, to separate genuine events from sensor faults and natural low-flow spikes; reported discharge locations, to test the bounded stretch; and verified responsible parties, to test the shortlist. For the Danube the ICPDR Accident Emergency Warning System and the European Pollutant Release and Transfer Register are candidate sources, though the latter is annual and constrains plausibility rather than timing.

Absent that evidence, a statistical outlier cannot be distinguished from a pollution event, nor proximity from responsibility. We therefore position this layer as triage that reduces an investigator's search space from an eleven-year multi-station record to a few dated, spatially bounded events – which the present evaluation does support – and claim no ability to identify polluters.

VI. Conclusion

This work built a two-stream spatial–temporal architecture in direct response to a Phase 1 finding: that a temporal model learns when water quality parameters move but not how far, because amplitude is governed by catchment conditions absent from station history. Adding a spatial expert over remotely sensed catchment features and fusing it under a learned attention gate improved R^2 on every parameter relative to the temporal stream alone, reaching 0.612 for nitrate-N, 0.459 for dissolved oxygen and 0.410 for conductance. The Phase 1 hypothesis is therefore supported.

The tuning comparison produced the study's most transferable result: substituting CMA-ES for a genetic algorithm as the spatial feature selector improved mean fused R^2 by 21% while reducing tuning time by 15%, outweighing the hyperparameter search strategy. For intercorrelated spectral features, feature selection appears the higher-leverage choice.

Two limitations bound these conclusions. The gate assigns 0.817 mean weight to the temporal stream despite the spatial expert being stronger for dissolved oxygen, where fusion scores 0.459 against 0.629 – leaving accuracy unclaimed and warranting a per-target rather than per-sample formulation. Total phosphorus stays unpredictable throughout, which we attribute to measurement noise near the detection limit rather than model capacity: a data problem no architecture change resolves.

A third limitation is evidential rather than technical: without confirmed incident records we cannot show that a flagged outlier is a pollution event, nor that a shortlisted facility discharged anything. We therefore scope this component as investigative triage rather than attribution.

Future work follows from all three. The gate should be trained with the streams frozen so it learns arbitration rather than co-adapting; per-target gating would let it defer to the spatial expert where that expert is demonstrably better. Extending the design so neighbouring stations inform one another, following the graph-based direction of Yan and Wang [2], would exploit the same channel geometry the screening layer already uses to bound its search region. Validating that layer will require linking flagged events to independent incident records, most plausibly as a small case-study series given how rarely confirmed events occur.

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